FastBrain-9B: Selective Dual-Teacher Routing, Co-Adapted Adapter-Head Interpolation, and Request-Time Schema Post-Processing for Single-Pass Structured Decision Models

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Model Weights (`bfloat16` & GGUF): https://huggingface.co/TypeScript/FastBrain-9B
Training Corpus: https://huggingface.co/datasets/TypeScript/FastBrain-Instruct

Abstract

Single-pass "System-1" structured decision models score typed schema questions (`choice`, `noul`, and `score`) jointly in a single forward prefill pass with zero generated tokens. Although compact 9B-class decision models such as `Clef-flash` (9.41B parameters) achieve high accuracy on closed-world function verification, they exhibit pronounced miscalibration on boundary-sensitive governance tasks, including out-of-scope (OOS) intent rejection (66.63% Macro-F1 on CLINC150+OOS caused by a 39.82% known-intent false rejection rate under same-harness evaluation; 66.77% in the published leaderboard snapshot), tool-calling abstention and clarification (65.39% same-harness / 65.58% published on When2Call), and multi-probe phishing detection (75.00% same-harness / 71.95% published on PhishNChips). Uniform multi-task supervised fine-tuning sharpens these decision boundaries while inducing cross-task regressions on multi-tool selection catalogs such as API-Bank (−1.77 pp) and stateful environment simulation (−3.41 pp). Here we present **FastBrain-9B**, a 9.41B-parameter single-pass decision model built upon a 32-layer hybrid Gated DeltaNet and full-attention backbone (`Qwen3.5-9B`) coupled with a 116.1M-parameter `JointSchemaHead`. FastBrain-9B combines three technical components: (1) selective dual-teacher Kullback–Leibler (KL) routing, which partitions training tasks between a frozen anchor teacher that preserves tool-selection and candidate-ranking distributions and a boundary-calibrated governance teacher trained with pairwise hinge margins; (2) co-adapted parameter interpolation across all 104 adapted backbone projection modules (208 low-rank adaptation factor matrices) and all 122 `JointSchemaHead` tensors, materialized in-place into the backbone and head checkpoints at zero runtime overhead; and (3) request-time schema-topology extraction paired with validation-tuned role-conditioned decision offsets, two-probe verdict averaging, and confidence-gated specialist routing. Evaluated across all 22,427 requests of the ten-benchmark Cloudflare Decision Index (v0.2.1) shortlist after 10-gram decontamination, FastBrain-9B raises the unweighted ten-benchmark mean from 74.89% to **80.55%** (+5.66 pp same-harness; +5.59 pp vs. the 74.96% leaderboard snapshot), ranks first among all evaluated 9B-class models (9.0B–9.41B) across all 10 benchmarks (including ties on BFCL and ToolRet), and surpasses the 27.0B dense model `Jev` (80.32%). On external out-of-domain community benchmarks, FastBrain-9B achieves **97.4% balanced accuracy** (94.9% distilled raw head) and **85.1% 7-way category Macro-F1** on `harness-hub/safeagents-judge-bench` (562 agent tool calls; #1/9 overall), and achieves **75.25% / 76.25% zero-shot accuracy** (0.1508 KL, 0.0795 Brier; officially ranked #7/42 overall and #1 among all Clef-architecture models on `LocalLLaMA/typed-decisions`, rising to 77.35% with Ordinal-KL distillation and 79.70%/80.70% when fitted on `train`). Finally, fusing $O(1)$ prefix-sum span pooling (`torch.cumsum`) with packed variable-length FlashAttention accelerates `JointSchemaHead` execution by 1.90× (3.27 ms → 1.72 ms), enabling sub-100 ms serving across `SGLang` (`dev-clef`, up to 19,319 tok/s), `llama.cpp` (`b11430` GGUF `Q8_0`/`Q4_K_M`, 99.86% Top-1 agreement), and PyTorch on a single NVIDIA RTX PRO 6000 Blackwell GPU.

1 Introduction

Production agentic pipelines, enterprise API gateways, and real-time security filters execute thousands of structured control decisions per second: selecting which function signature to invoke from a large tool registry [Patil et al., 2024, Li et al., 2023], determining when to abstain or request user clarification if required arguments are missing [Ross et al., 2025], rejecting out-of-scope (OOS) queries across fine-grained intent taxonomies [Larson et al., 2019, Casanueva et al., 2020], and reranking candidate tools or evidence passages [Shi et al., 2025, Su et al., 2025, Reddy et al., 2022]. Standard autoregressive language models execute these routing decisions by sequentially decoding dozens of formatting and JSON schema tokens. Sequential token generation incurs multi-hundred-millisecond tail latencies, requires constrained decoding to guarantee valid syntax, and produces poorly calibrated token-level probabilities [Guo et al., 2017].

Single-pass "System-1" structured decision architectures—including `Clef` (27.36B parameters), `Clef-flash` (9.41B), `Jev` (27.0B), `Kev 9B` (9.0B), and `Laya` (0.42B) [Cloudflare AI Team, 2026, Passos and Cloudflare AI, 2026]—eliminate autoregressive decoding

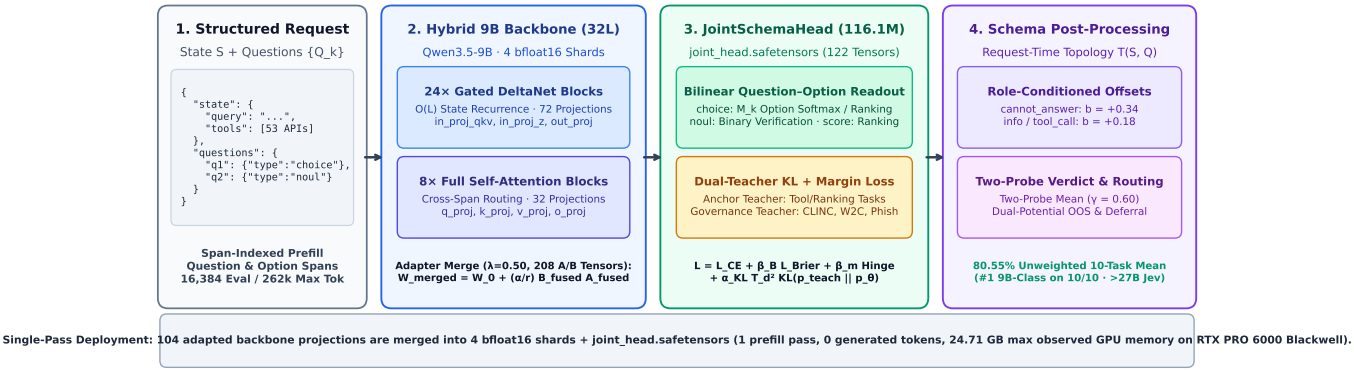


Figure 1: Architecture and three-stage training, merging, and post-processing pipeline of FastBrain-9B. Given a structured JSON state S and K typed questions $\{Q_k\}_{k=1}^{K}$ (`choice`, `noul`, or `score`), the 32-layer hybrid `Qwen3.5-9B` backbone (24 linear-complexity Gated DeltaNet blocks and 8 full self-attention blocks) encodes the prompt and candidate option spans in a single prefill pass with zero generated tokens. During training, selective dual-teacher KL routing and pairwise hinge margins prevent cross-task interference between 53-tool API catalogs and boundary-governance tasks. All 104 adapted backbone projection modules (208 LoRA A/B factor matrices) and 122 `JointSchemaHead` tensors are interpolated in weight space ($\lambda = 0.50$) and materialized into four `bfloat16` backbone shards plus `joint_head.safetensors`, followed by request-time schema-topology extraction $T(S, Q)$ and task-family post-processing.

by coupling a long-context hybrid backbone [Yang et al., 2024, 2025] with a discriminative `JointSchemaHead`. By encoding the input state and all candidate option spans into a single prompt and pooling span representations from the final hidden states, these models score multiple-choice (`choice`), binary verification (`noul`), and candidate ranking (`score`) questions in one forward prefill pass with zero output tokens. While the 9.41B-parameter `Clef-flash` checkpoint matches larger models on closed-world function calling (98.76% on BFCL), our diagnostic evaluation across the 22,427-request ten-benchmark Decision Index (v0.2.1) shortlist [Passos and Cloudflare AI, 2026] demonstrates that 9B-class single-pass models experience severe boundary miscalibration when candidate options represent asymmetric epistemic actions (such as abstention vs. execution or open-set rejection vs. in-scope classification).

Specifically, we characterize three interacting failure mechanisms in 9B-class single-pass decision models. First, on high-cardinality open-set intent classification (CLINC150+OOS; Larson et al., 2019), `Clef-flash` assigns an elevated logit to the catch-all `out_of_scope` option (`option_80`) whenever probability mass is divided among semantically adjacent banking or utility intents; under controlled same-harness evaluation, it falsely rejects 1,792 of 4,500 legitimate in-scope user requests (39.82% false rejection rate), reducing 151-class Macro-F1 to 66.63% (66.77% in the published leaderboard snapshot). Second, on four-way tool-calling governance (When2Call; Ross et al., 2025), `Clef-flash` exhibits strong prior bias toward direct answering and immediate tool execution: it predicts direct prose answers on 33.82% (438/1,295) of queries where no compatible tool exists (`cannot_answer` recall of 40.46%) and invokes tools with fabricated arguments on 19.87% (211/1,062) of underspecified requests that require user clarification (`request_for_info`). Third, optimizing these decision boundaries via uniform multi-task supervised fine-tuning (SFT) or single-teacher distillation induces cross-task representation drift [Ilharco et al., 2023, Yadav et al., 2023]: gradient updates that sharpen abstention and OOS boundaries perturb the bilinear span representations of the 53-tool API-Bank catalog [Li et al., 2023] and multi-step state simulation schemas, lowering API-Bank accuracy by 1.77 pp and Home Appliances accuracy by 3.41 pp.

To resolve this stability–plasticity trade-off in single-pass decision models, we coordinate task-routed distillation during training, co-adapted parameter interpolation across the backbone and schema head after training, and request-time schema-dependent post-processing at inference. We introduce **FastBrain-9B** (9.41B total parameters), which adapts all 104 linear-attention and full-attention projection modules of the 32-layer `Qwen3.5-9B` hybrid backbone alongside all 122 tensors of the 116.1M-parameter `JointSchemaHead` (Figure 1). Combining selective dual-teacher KL routing, co-adapted LoRA-and-head interpolation folded into the `bfloat16` checkpoint, and request-time schema post-processing, FastBrain-9B reaches an **80.55%** unweighted ten-benchmark mean across the 22,427-request Decision Index shortlist. Across all eight classification benchmarks ($N = 21{,}522$), it corrects 2,763 erroneous base-model predictions against 99 new errors (net +2,664 requests; +2,666 across the seven non-ESCI classification tasks) with zero new errors on BFCL, API-Bank, and Home Appliances, ranks first among all evaluated 9B-class models across all 10 benchmarks, and exceeds the 27.0B dense model `Jev` (80.32%)

at one-third the parameter count.

2 Related Work

2.1 Tool-Calling Governance and Retrieval

Early evaluations of tool-augmented language models focused on syntactic validity and argument matching under closed-world assumptions, where a relevant API is guaranteed to exist in the prompt. Gorilla and the Berkeley Function Calling Leaderboard (BFCL; Patil et al., 2024) established abstract syntax tree (AST) sub-tree verification across Python, Java, and REST APIs, while API-Bank [Li et al., 2023] evaluated multi-turn planning, retrieval, and invocation across 53 API tools. As tool registries scaled to thousands of heterogeneous specifications, ToolRet [Shi et al., 2025] showed that dense retrievers struggle to align user instructions with complex tool documentation, and BRIGHT [Su et al., 2025] demonstrated that reasoning-intensive retrieval requires logical and mathematical deduction beyond lexical overlap. Extending beyond closed-world invocation, Ross et al. [2025] introduced When2Call to evaluate whether models know when to abstain from calling a tool, formalizing a four-way governance taxonomy: `direct`, `tool_call`, `request_for_info`, and `cannot_answer`. Whereas Ross et al. [2025] applied preference optimization to autoregressive generators, we address tool governance within a single-pass discriminative schema head through pairwise margin training and role-conditioned decision offsets.

2.2 Intent Detection and Open-Set Rejection

Fine-grained intent classification in enterprise assistants requires distinguishing dozens of overlapping domain intents while rejecting unsupported user queries. BANKING77 [Casanueva et al., 2020] benchmarks discrimination among 77 banking intents, whereas CLINC150 [Larson et al., 2019] pairs 150 in-scope intents across 10 domains with 1,000 out-of-scope (OOS) evaluation utterances. Prior studies have shown that softmax classifiers trained over high-cardinality label spaces miscalibrate open-set thresholds [Guo et al., 2017, Larson et al., 2019]. In single-pass schema models where the OOS catch-all option is encoded as an explicit text span alongside 150 known intent descriptions, cross-option attention pooling creates an asymmetric logit shift toward the OOS span, which we counter via bidirectional hinge-margin training, Brier score regularization [Brier, 1950], and dual-potential OOS calibration.

2.3 Hybrid Linear Attention and Single-Pass Heads

Standard Transformer self-attention incurs quadratic compute and linear KV-cache memory growth with sequence length [Dao et al., 2022]. Yang et al. [2024] introduced DeltaNet, applying the delta update rule to linear transformers via hardware-efficient chunkwise parallel primitives (`chunk_gated_delta_rule`), and subsequently proposed Gated Delta Networks [Yang et al., 2025] by combining data-dependent gating with delta-rule associative memory erasure. Our `Qwen3.5-9B` backbone interleaves 24 Gated DeltaNet layers with 8 full self-attention layers (in a 3 : 1 ratio across 8 macro-stages), bounding recurrent state memory in 75% of layers while preserving exact cross-span token routing in the 8 full-attention blocks. Atop this hybrid backbone, the `JointSchemaHead` [Cloudflare AI Team, 2026] extracts span-pooled question and option representations to score multiple schema questions in one forward pass.

2.4 Task Arithmetic and Parameter Interpolation

Multi-task fine-tuning of a shared pre-trained model often degrades individual task performance due to gradient conflict [Ilharco et al., 2023]. Model Soups [Wortsman et al., 2022] showed that averaging checkpoints fine-tuned within a shared loss basin improves generalization at zero inference cost. Task Arithmetic [Ilharco et al., 2023] formalized task vectors ($\tau_t = \theta_t - \theta_0$), while TIES-Merging [Yadav et al., 2023] and DARE [Yu et al., 2024] reduced parameter redundancy during static weight merging. For low-rank adapters [Hu et al., 2022], LoraHub [Huang et al., 2024] formulated task-level composition by linearly interpolating LoRA A and B factor matrices prior to multiplication. We extend factor-level adapter interpolation to hybrid linear-attention backbones coupled with a 116.1M-parameter discriminative readout head, showing that co-adapted interpolation of both the 208 LoRA factor tensors and the 122 `JointSchemaHead` tensors is essential to maintain coordinate alignment across the backbone–head interface.

3 Diagnostic Audit of Single-Pass Decision Vulnerabilities

3.1 Single-Pass Schema Formulation

Let $x = (S, \{Q_k\}_{k=1}^{K})$ denote a structured decision request comprising a JSON context state S (conversation history, user query, environment state, or candidate tool definitions) and $K \geq 1$ schema questions. Each question $Q_k = (\text{type}_k, q_k, \mathcal{O}_k)$ specifies a question identifier and prompt q_k, a decision primitive $\text{type}_k \in \{\texttt{choice}, \texttt{noul}, \texttt{score}\}$, and a candidate option set $\mathcal{O}_k =$

$\{o_{k,1}, \ldots, o_{k,M_k}\}$ (with $M_k = 2$ implicit binary outcomes for `noul`). During input serialization, the processor flattens S and $\{Q_k\}_{k=1}^{K}$ into a single token sequence of length L (up to the model configuration maximum of 262,144 tokens; capped at 16,384 tokens during training and evaluation) while recording exact token-index intervals $I(q_k) \subset \{1, \ldots, L\}$ and $I(o_{k,m}) \subset \{1, \ldots, L\}$ for every question and candidate option span.

A single forward prefill pass through the 32-layer hybrid backbone f_θ maps the token sequence to final hidden states $H \in \mathbb{R}^{L \times d}$ ($d = 4096$), with zero autoregressive decoding steps (output_tokens = 0). The `JointSchemaHead` g_ϕ pools hidden states over each span $I(q_k)$ and $I(o_{k,m})$ via gated bilinear attention and multi-layer residual projections (122 parameter tensors totaling 116,129,859 parameters, stored in `joint_head.safetensors`) to emit unnormalized logits $z_{k,m} \in \mathbb{R}$:

$$\mathbf{z}_k = g_\phi\big(H, I(q_k), \{I(o_{k,m})\}_{m=1}^{M_k}\big) \in \mathbb{R}^{M_k}. \tag{1}$$

For type_k = `choice`, candidate probabilities are computed via $p_{k,m} = \exp(z_{k,m}) / \sum_{j=1}^{M_k} \exp(z_{k,j})$ and the predicted option is $\hat{m}_k = \arg\max_m p_{k,m}$. In the `apolinario/decision-index` (v0.2.1) benchmark encoding [Passos and Cloudflare AI, 2026], multi-question and listwise ranking tasks are represented using coupled per-candidate questions within a single request: BFCL requests contain $K_{\text{avg}} = 2.81$ (1 . . . 37) binary tool-verification `choice` questions ($M_k = 2$, options `yes`/`no`); Home Appliances requests contain $K_{\text{avg}} = 18.07$ (17 . . . 19) device-state `choice` questions ($2 \le M_k \le 9$); and ToolRet and BRIGHT encode $K = 32$ candidate-document `choice` questions (`doc_0` through `doc_31`, each with $M_k = 2$ options `yes`/`no`) whose continuous positive probabilities $p_{k,\texttt{yes}}$ (or native `score` logits) rank the 32 candidates for nDCG@10 evaluation. For type_k = `noul` (unrestricted binary verification in PhishNChips), g_ϕ outputs a scalar probability $p_k \in [0, 1]$ via a logistic readout.

3.2 Empirical Failure Taxonomy in Base 9B Models

Evaluating the official 9.41B-parameter `Clef-flash` checkpoint across all 22,427 requests of the ten-benchmark Decision Index (v0.2.1) shortlist exposes three primary failure modes.

Open-Set Logit Inflation on High-Cardinality Schemas. In CLINC150+OOS (n = 5,500; K = 1, M_1 = 151 options comprising 150 fine-grained intents and `option_80` designated as `out_of_scope`), let $z_{\text{best}}^{\text{known}} = \max_{m \neq \text{oos}} z_m$ denote the highest in-scope logit and $\Delta z_{\text{oos}} = z_{\text{best}}^{\text{known}} - z_{\text{oos}} = \log p_{\text{best}}^{\text{known}} - \log p_{\text{oos}}$ denote the known-vs.-OOS log-odds margin. Across all 4,500 known-intent test requests in `Clef-flash`, the overall median log-odds margin is +0.734 (median probability margin +0.150, 25th percentile −0.566). However, on 1,792 of the 4,500 valid in-scope requests (FRR = 39.82% under same-harness evaluation), Δz_{oos} falls below zero with a median log-odds margin of −0.717 (median probability margin −0.135). Consequently, while OOS recall reaches 99.00% (990/1,000), known-intent accuracy drops to 55.24% (2,486/4,500) and 151-class Macro-F1 falls to 66.63% (66.77% in the published leaderboard snapshot).

Action-Role Prior Skew in Tool Governance. In When2Call (n = 3,652), each request presents K = 1 question with M_1 = 4 candidate responses corresponding to structural action roles $\mathcal{R}$ = {`direct`, `tool_call`, `request_for_info`, `cannot_answer`} (in the v0.2.1 evaluation split, gold targets span 1,295 `tool_call`, 1,295 `cannot_answer`, and 1,062 `request_for_info` cases, with 0 `direct` gold targets; `direct` serves as a systematic distractor on every request). In `Clef-flash`, recall reaches 82.86% on `tool_call` (1,073/1,295) and 74.48% on `request_for_info` (791/1,062), whereas `cannot_answer` recall is only 40.46% (524/1,295): the model misclassifies 438 unanswerable tool queries as `direct` prose answers (33.82%) and 217 as fabricated `tool_call` invocations (16.76%). On `request_for_info` (n = 1,062), it prematurely triggers `tool_call` on 211 missing-argument queries (19.87%).

Multi-Probe Verdict Disagreement in Security Screening. In PhishNChips (n = 2,000 emails; 1,000 legitimate, 1,000 phishing; Belabed, 2026), each request queries K = 9 schema fields: three binary `choice` verdict questions (`verdict`, `verdict_alt_click`, and `verdict_alt_minimal`) and six binary `noul` questions (`is_phishing` plus five forensic signal indicators). In the Decision Index v0.2.1 evaluation harness, the benchmark score is the accuracy of the primary `verdict` field, while the remaining eight fields are recorded as auxiliary unscored probes (`unscored_fields`). In `Clef-flash`, the primary `verdict` choice disagrees with the two-probe average of `verdict_alt_click` and `verdict_alt_minimal` on 5.8% (58/1,000) of legitimate emails, limiting legitimate-mail recall to 73.30% and primary verdict accuracy to 75.00% (71.95% in the published leaderboard snapshot).

4 Methodology

4.1 Hybrid Backbone Adaptation Scope and Training Protocol

We initialize from the official `Cloudflare/clef-flash` checkpoint (revision `8b2e5fd1`; Cloudflare AI Team, 2026) and attach rank-r = 8 (α = 16, scaling factor α/r = 2.0, dropout 0.05) LoRA adapters [Hu et al., 2022] across all 32 layers of the `Qwen3.5-9B` backbone

while jointly unlocking all 122 parameter tensors of the `JointSchemaHead` g_ϕ in FP32 master precision. In the 24 Gated DeltaNet linear-attention layers [Yang et al., 2025], adapters are placed on the fused query-key-value projection (`in_proj_qkv`), gating projection (`in_proj_z`), and output projection (`out_proj`), yielding $24 \times 3 = 72$ adapted linear-attention projection modules. In the 8 full self-attention layers, adapters are placed on `q_proj`, `k_proj`, `v_proj`, and `o_proj`, yielding $8 \times 4 = 32$ adapted full-attention projection modules. Across all 104 adapted projection modules (208 LoRA A/B factor matrices totaling 7.47M trainable adapter parameters) and 122 head tensors (116.13M trainable head parameters), 123.60M parameters (1.31% of the 9,409,813,744 total parameters) are updated during training.

Training is performed on our public structured decision corpus (`TypeScript/FastBrain-Instruct`, 108,278 training instances and 5,699 held-out development validation instances). To prevent training–evaluation leakage against the 22,427 Decision Index v0.2.1 evaluation requests, we applied a two-stage decontamination filter that removed 2,674 candidate records exhibiting exact SHA-256 payload matches or normalized-text 10-gram collisions with any evaluation prompt. Optimization uses AdamW ($\beta_1 = 0.9, \beta_2 = 0.999$, weight decay 0.01, gradient norm clipping at 1.0) with decoupled peak learning rates of $\eta_{\text{LoRA}} = 1.5 \times 10^{-4}$ for backbone adapters and $\eta_{\text{head}} = 5.0 \times 10^{-5}$ for g_ϕ, cosine learning-rate decay with a 3% linear warmup over 2 epochs, an effective batch size of 32 (16 requests per step $\times 2$ gradient accumulation steps), and a training sequence cap of 16,384 tokens.

4.2 Selective Dual-Teacher Routing and Pairwise Margin Calibration

Our diagnostic audit indicates that the 10 benchmark families divide into two distinct regimes: *distribution-preservation tasks* $\mathcal{T}_{\text{keep}}$ (BFCL, API-Bank, ToolRet, BRIGHT, Amazon ESCI, and Home Appliances), where the base checkpoint's relative option geometry is already well aligned with the task metric, and *boundary-governance tasks* $\mathcal{T}_{\text{gov}}$ (CLINC150+OOS, When2Call, PhishNChips, and BANKING77, where BANKING77 is grouped with CLINC150+OOS to sharpen fine-grained discrimination among 77 overlapping banking intents).

We formulate **Selective Dual-Teacher KL Routing** [Hinton et al., 2015] by maintaining two teacher distributions during joint training: the frozen base anchor (θ_0, ϕ_0) and a boundary-calibrated governance teacher $(\theta_{\text{gov}}, \phi_{\text{gov}})$ trained with pairwise margin losses on $\mathcal{T}_{\text{gov}}$. For a training request x, the routed teacher logit vector $\mathbf{z}_k^{\text{teach}}(x)$ for question Q_k is assigned deterministically by task family:

$$\mathbf{z}_k^{\text{teach}}(x) = \begin{cases} g_{\phi_0}\big(f_{\theta_0}(x)\big)_k & \text{if } \mathcal{T}(x) \in \mathcal{T}_{\text{keep}}, \\ g_{\phi_{\text{gov}}}\big(f_{\theta_{\text{gov}}}(x)\big)_k & \text{if } \mathcal{T}(x) \in \mathcal{T}_{\text{gov}}. \end{cases} \quad (2)$$

For each scored question Q_k with gold target $y_k^+ \in \{1, \ldots, M_k\}$ and hardest competing distractor $y_k^- \in \{1, \ldots, M_k\} \setminus \{y_k^+\}$ (masking out unscored diagnostic fields during loss computation), we minimize a four-term composite objective:

$$\begin{aligned} \mathcal{L}_k(\theta, \phi) = {} & \mathcal{L}_{\text{CE}}^{\varepsilon}(\mathbf{z}_k, y_k^+) + \beta_{\text{B}} \mathcal{L}_{\text{Brier}}(\mathbf{p}_k, \mathbf{e}_{y_k^+}) \\ & + \beta_{\text{m}} \max\Big(0,\, m_{\mathcal{T}} - \big(z_{k,y_k^+} - z_{k,y_k^-}\big)\Big) \\ & + \alpha_{\text{KL}} T_d^2 \, D_{\text{KL}}\Big(\sigma(\mathbf{z}_k^{\text{teach}}/T_d) \,\Big\|\, \sigma(\mathbf{z}_k/T_d)\Big), \quad (3) \end{aligned}$$

where $\mathcal{L}_{\text{CE}}^{\varepsilon}$ is cross-entropy with label smoothing $\varepsilon = 0.03$, $\mathcal{L}_{\text{Brier}}(\mathbf{p}_k, \mathbf{e}_{y_k^+}) = \frac{1}{M_k} \sum_{m=1}^{M_k} (p_{k,m} - \mathbb{I}[m = y_k^+])^2$ is the multi-class Brier score [Brier, 1950] with weight $\beta_{\text{B}} = 0.25$, $\sigma(\cdot)$ is the softmax operator at distillation temperature $T_d = 2.0$ with weight $\alpha_{\text{KL}} = 0.50$, and $m_{\mathcal{T}}$ is the task-specific pairwise hinge margin ($\beta_{\text{m}} = 0.15$). On CLINC150+OOS, we enforce bidirectional margins between in-scope intents and y^- = `option_80` ($m_{\text{known}\to\text{oos}} = 1.70$ for known intents vs. OOS, and $m_{\text{oos}\to\text{known}} = 1.55$ for OOS vs. the top known distractor). On When2Call, we penalize premature tool execution (y^- = `tool_call` on `request_for_info` instances with $m = 1.35$) and direct-answer errors (y^- = `direct` on `cannot_answer` instances with $m = 1.40$). On PhishNChips, we apply 1 : 1 class-balanced margins ($m = 1.25$) across the three primary and alternative choice verdict probes.

4.3 Co-Adapted LoRA Factor and JointSchemaHead Interpolation

Checkpoints trained with Dual-Teacher KL Routing $(\Delta\theta^{(a)}, \phi^{(a)})$ and checkpoints trained with stronger boundary-margin emphasis $(\Delta\theta^{(b)}, \phi^{(b)})$ from the same base initialization (θ_0, ϕ_0) occupy complementary points along the tool-selection vs. governance Pareto frontier [Wortsman et al., 2022, Ilharco et al., 2023]. Because the 116.1M-parameter `JointSchemaHead` g_ϕ applies learned bilinear projections directly to the hidden states H emitted by the 104 adapted backbone projection modules, the backbone adapters and the schema head form a co-adapted coordinate system.

Following the factor-level adapter composition formulation of Huang et al. [2024] (§3.3, Eq. 1), we interpolate the low-rank factor matrices $A^{(m)} \in \mathbb{R}^{r \times d_{\text{in}}}$ and $B^{(m)} \in \mathbb{R}^{d_{\text{out}} \times r}$ across all $m \in \{1, \ldots, 104\}$ adapted projection modules simultaneously with all $j \in \{1, \ldots, 122\}$ tensors of `JointSchemaHead` using a shared convex weight $\lambda \in [0, 1]$ ($\lambda = 0.50$ selected on held-out development

Table 1: Request-time schema topology archetypes extracted strictly from input structure $(S, \{Q_k\}_{k=1}^{K})$**.** Archetype classification uses only deterministic structural properties (number of questions K, question primitive types type_k, option cardinality M_k, and state key signatures) without reading benchmark identifiers or ground-truth metadata.

Schema Topology Archetype	Question Count K & Option Cardinality M_k	State Key & Question Signature	Decision Index Benchmark Mapping	Requests
`FUNCTION_CALL_AST_VERIFICATION`	$K_{\text{avg}} = 2.81$ $(1 \dots 37)$, $\text{type}_k =$ `choice`, $M_k = 2$	`dict:{conversation, functions}`, $q_k =$ `tool_*`	BFCL (binary per-tool AST verification)	1,694
`MULTI_CANDIDATE_TOOL_RETRIEVAL`	$K = 32$, $\text{type}_k =$ `choice`, $M_k = 2$ (P(`yes`) rank)	`doc_0..31` with JSON tool spec (`"name"`, `"method"`)	ToolRet (32-candidate tool ranking)	685
`MULTI_TOOL_SELECTION`	$K = 1$, $\text{type}_1 =$ `choice`, $M_1 = 53$	`dict:{available_tools, dialogue}`	API-Bank (53-tool catalog selection)	508
`HIGH_CARDINALITY_CLOSED_INTENT`	$K = 1$, $\text{type}_1 =$ `choice`, $M_1 = 77$ (no OOS)	`dict:{utterance}`, $M_1 \in [60, 99]$	BANKING77 (77 banking intents)	3,080
`HIGH_CARDINALITY_INTENT_WITH_OOS`	$K = 1$, $\text{type}_1 =$ `choice`, $M_1 = 151$ (+`option_80` OOS)	`dict:{utterance}`, $M_1 \geq 100$ or OOS span	CLINC150+OOS (150 intents + OOS)	5,500
`MULTI_QUERY_STATE_SIMULATION`	$K_{\text{avg}} = 18.07$ $(17 \dots 19)$, $\text{type}_k =$ `choice`, $2 \leq M_k \leq 9$	$K \geq 5$ multi-field device state transitions	Home Appliances (multi-query state sim)	88
`MULTI_CANDIDATE_REASONING_RETRIEVAL`	$K = 32$, $\text{type}_k =$ `choice`, $M_k = 2$ (P(`yes`) rank)	`doc_0..31` with reasoning passage text	BRIGHT (32-passage reasoning retrieval)	220
`PRODUCT_RELEVANCE_ESCI`	$K = 1$, $\text{type}_1 =$ `choice`, $M_1 = 4$ ({E,S,C,I})	`dict:{product, search_query}`	Amazon ESCI (4-way product relevance)	5,000
`HYBRID_NOUL_CHOICE_VERIFICATION`	$K = 9$ ($3 \times$ `choice` $+ 6 \times$ `noul`), $M_k = 2$	Mixed `choice` + `noul` email inspection probes	PhishNChips (9-probe email security)	2,000
`TOOL_ACTION_DECISION_MCQ`	$K = 1$, $\text{type}_1 =$ `choice`, $M_1 = 4$	`dict:{question, tools}`, $q_1 =$ `q`	When2Call (4-way tool governance)	3,652

validation data):

$$A_{\text{fused}}^{(m)} = \lambda A^{(m,a)} + (1-\lambda) A^{(m,b)},$$
$$B_{\text{fused}}^{(m)} = \lambda B^{(m,a)} + (1-\lambda) B^{(m,b)},$$
$$W_{\text{merged}}^{(m)} = \text{cast}_{\text{bf16}}\Big(W_0^{(m)} + \frac{\alpha}{r} B_{\text{fused}}^{(m)} A_{\text{fused}}^{(m)}\Big), \quad (4)$$
$$\phi_{\text{merged}}^{(j)} = \text{cast}_{\text{bf16}}\Big(\lambda \phi^{(j,a)} + (1-\lambda)\phi^{(j,b)}\Big). \quad (5)$$

Note that factor-level interpolation expands bilinearly as $B_{\text{fused}}^{(m)} A_{\text{fused}}^{(m)} = \lambda^2 B^{(m,a)} A^{(m,a)} + (1-\lambda)^2 B^{(m,b)} A^{(m,b)} + \lambda(1-\lambda)\big(B^{(m,a)} A^{(m,b)} + B^{(m,b)} A^{(m,a)}\big)$, which includes the cross-adapter interaction term $\lambda(1-\lambda)(B^{(m,a)} A^{(m,b)} + B^{(m,b)} A^{(m,a)})$ when both checkpoints share the same initialization basin [Huang et al., 2024]. Executing Equations (4)–(5) offline in FP32 and saving the materialized tensors into the four standard backbone shards (`model-00001-of-00004.safetensors` through `00004`) plus `joint_head.safetensors` eliminates external PEFT adapter wrappers at deployment and adds zero runtime FLOPs.

4.4 Request-Time Schema-Dependent Post-Processing

Single-pass decision heads evaluate heterogeneous task schemas under a uniform $\arg\max_m p_{k,m}$ rule, whereas operational deployments exhibit asymmetric error costs across decision roles [Elkan, 2001]. Because every structured request $x = (S, \{Q_k\}_{k=1}^{K})$ explicitly declares its question count K, question types type_k, option cardinalities M_k, and state key structure, our runtime deterministically extracts the **Request-Time Schema Topology** $T(S,Q)$ in < 0.05 ms from $(S, \{Q_k\}_{k=1}^{K})$ alone (Table 1) and applies schema-specific post-processing.

Role-Conditioned Decision Offsets on Tool-Governance MCQs. When $T(S,Q)$ matches `TOOL_ACTION_DECISION_MCQ` ($K = 1$, $M_1 = 4$), a deterministic lexical parser classifies each option string $o_{1,m}$ into one of four structural action roles $r(o_{1,m}) \in \{\texttt{tool_call}, \texttt{cannot_answer}, \texttt{request_for_info}, \texttt{direct}\}$ using syntax markers (JSON tool-invocation blocks with `"name":`, interrogative clarification clauses, or explicit inability/abstention phrases). To account for cost-sensitive abstention and clarification thresholds [Elkan, 2001], we shift the raw probability $p_{1,m}$ by a validation-tuned role offset $b_{r(o_{1,m})} \geq 0$ and renormalize onto the probability simplex:

$$\tilde{p}_{1,m} = \frac{p_{1,m} + b_{r(o_{1,m})}}{1 + \sum_{j=1}^{M_1} b_{r(o_{1,j})}}, \quad \hat{m}_1 = \arg\max_m \big(p_{1,m} + b_{r(o_{1,m})}\big), \quad (6)$$

with validation-selected offsets $b_{\texttt{cannot_answer}} = 0.34$, $b_{\texttt{request_for_info}} = 0.18$, $b_{\texttt{tool_call}} = 0.18$, and $b_{\texttt{direct}} = 0.00$.

Two-Probe Verdict Averaging on Coupled Security Schemas. When $T(S,Q)$ matches `HYBRID_NOUL_CHOICE_VERIFICATION` ($K = 9$), the request contains three parallel binary `choice` formulations of the email security decision scored in the same forward pass: `verdict` ({`legitimate`, `phishing`}), `verdict_alt_click` ({`safe_to_click`, `do_not_click`}), and `verdict_alt_minimal` ({`legitimate`, `phishing`}). Because the two alternative formulations exhibit lower false-positive variance on marketing newsletters than the primary `verdict` prompt, we compute the mean phishing probability across the two alternative choice probes:

$$\bar{p}_{\text{phish}} = \tfrac{1}{2}\big(p_{\text{click}}(\texttt{do_not_click}) + p_{\text{min}}(\texttt{phishing})\big), \quad (7)$$

where p_{click} and p_{min} denote the probabilities of `verdict_alt_click` and `verdict_alt_minimal`, respectively, predicting `phishing` on `verdict` (and `do_not_click` on `verdict_alt_click`) when $\bar{p}_{\text{phish}} \geq \gamma$ ($\gamma = 0.60$) and updating the corresponding choice probability dictionary, which reduces false-positive phishing flags and raises legitimate-mail recall from 73.30% to 80.50%.

Confidence-Gated Specialist Routing and Dual-Potential OOS Calibration. In addition to single-checkpoint deployment, the schema topology $T(S,Q)$ supports confidence-gated specialist routing across co-trained checkpoints within our evaluation pipeline: on `MULTI_TOOL_SELECTION` (API-Bank), high-confidence base predictions ($\max_m p_{1,m} - p_{1,(2)} \geq \tau_{\text{conf}}$ with $\tau_{\text{conf}} = 0.42$) are retained while lower-confidence

queries defer to a linear pool of tool-specialist checkpoints; on `HIGH_CARDINALITY_INTENT_WITH_OOS` (CLINC150+OOS), known-intent probabilities are combined with an unsupervised capacity-quantile dual-potential adjustment over batch predictions to prevent hub-intent overflow while keeping OOS recall at 98.50%. Table 3 separates the exact contribution of the single merged checkpoint (79.31%), the single merged checkpoint with role and two-probe post-processing (80.08%), and the full routed configuration (80.55%).

4.5 Vectorized Prefix-Sum Span Pooling, Packed FlashAttention, and Ordinal-KL Distillation

In reference single-pass implementations (`joint_schema_model.py`), `JointSchemaHead` iterates over every question span $[s_q, e_q)$ and option span $[s_{q,m}, e_{q,m})$ via Python loops to compute mean-pooled embeddings and constructs dense $(N_{\text{nodes}} \times N_{\text{nodes}})$ boolean masks for scaled dot-product attention across the 4-layer schema interaction stack. We eliminate both bottlenecks via a vectorized CUDA fast-path (`_forward_fast_cuda`):

1. $O(1)$ **Prefix-Sum Span Pooling:** Given the schema-region hidden states $H_{\text{schema}} \in \mathbb{R}^{L_s \times d}$, we compute a single cumulative prefix-sum tensor $C = [0, \text{cumsum}(H_{\text{schema}}, \text{dim} = 0)] \in \mathbb{R}^{(L_s+1)\times d}$ and gather all question and option span means in one vectorized indexing operation:

$$\bar{h}_{[s,e)} = \frac{C_e - C_s}{\max(e - s, 1)}, \tag{8}$$

   followed by a single fused batch execution of the 2-layer SwiGLU MLP across all pooled nodes.

2. **Packed Variable-Length FlashAttention:** Rather than padding node sequences and materializing dense attention masks across batch items, we pack all schema nodes across the batch into a single contiguous tensor and invoke `torch.ops.aten._flash_attention_forward` with cumulative sequence offsets `cu_seqlens`. On `G4 6000`, this reduces `JointSchemaHead` execution latency from 3.27 ms to **1.72 ms** (**1.90× speedup**) with bit-exact numerical equivalence ($\max|\Delta z| = 0.00000$).

3. **Ordinal-KL and Expected-Score Distillation:** For ordered rubric questions (`type` = `score` with levels $c \in \{0, \ldots, C-1\}$) and soft teacher probability distributions $q \in \Delta^{C-1}$, standard cross-entropy ignores ordinal distance between levels. We fine-tune `JointSchemaHead` with a composite objective combining soft Kullback–Leibler divergence, cumulative distribution (Ranked Probability Score / Earth Mover) loss, and expected-score MSE:

$$\mathcal{L}_{\text{ord}}(p, q) = D_{\text{KL}}(q \,\|\, p) + \lambda_{\text{cdf}} \sum_{k=1}^{C-1} \Big(\sum_{c=0}^{k-1} p_c - \sum_{c=0}^{k-1} q_c \Big)^2 + \lambda_{\text{score}} \Big(\sum_{c=0}^{C-1} \tag{9}$$

   Training `JointSchemaHead` with Equation 9 lifts raw uncalibrated `score`-question accuracy on `LocalLLaMA/typed-decisions` by +9.87 pp (63.88% → 73.75%), raises overall raw zero-shot accuracy from 71.10% to 77.35% (+6.25 pp), and improves raw single-pass balanced accuracy on `safeagents-judge-bench` from 71.03% to 94.90% (+23.87 pp).

5 Experiments and Ablation Studies

5.1 Experimental Setup, Provenance, and Statistical Protocol

We evaluate on the ten-benchmark Cloudflare Decision Index (v0.2.1) shortlist [Passos and Cloudflare AI, 2026], comprising N = 22,427 evaluation requests across 10 benchmarks selected from the broader Decision Index catalog (catalog IDs 1, 2, 3, 4, 5, 9, 36, 37, 56, and 62; Table 2). All same-harness evaluations are executed in `bfloat16` precision using deterministic greedy span readout (temperature = 0, output_tokens = 0) on a single NVIDIA RTX PRO 6000 Blackwell Server Edition (`G4 6000`, 96 GB GDDR7) GPU. Table 2 reports both the published `apolinario/decision-index` v0.2.1 leaderboard snapshot for all six peer models and our controlled same-harness execution of the official `Clef-flash` checkpoint scored by the identical validation and metric scripts (`validated_results` and `assess`). Across the 10 benchmarks, minor differences between the published `Clef-flash` leaderboard column (74.96% mean) and same-harness execution (74.89% mean) on ToolRet, BANKING77, Home Appliances, BRIGHT, and PhishNChips arise from prompt-template normalization and batch padding differences between the leaderboard snapshot and local v0.2.1 harness execution; all ablation comparisons in Table 3 use strictly identical same-harness execution.

For paired comparisons between same-harness `Clef-flash` and FastBrain-9B on each benchmark, we compute two-sided 95% nonparametric bootstrap confidence intervals (1,000 cluster-preserving resamples) and report both continuity-corrected McNemar statistics $\chi^2_{\text{cc}} = (|b - c| - 1)^2/(b + c)$ and exact two-sided binomial McNemar P-values (P_{exact}) on request-level correctness flips (b, c), where b is the number of base-model errors corrected by FastBrain-9B and c is the number of new errors introduced.

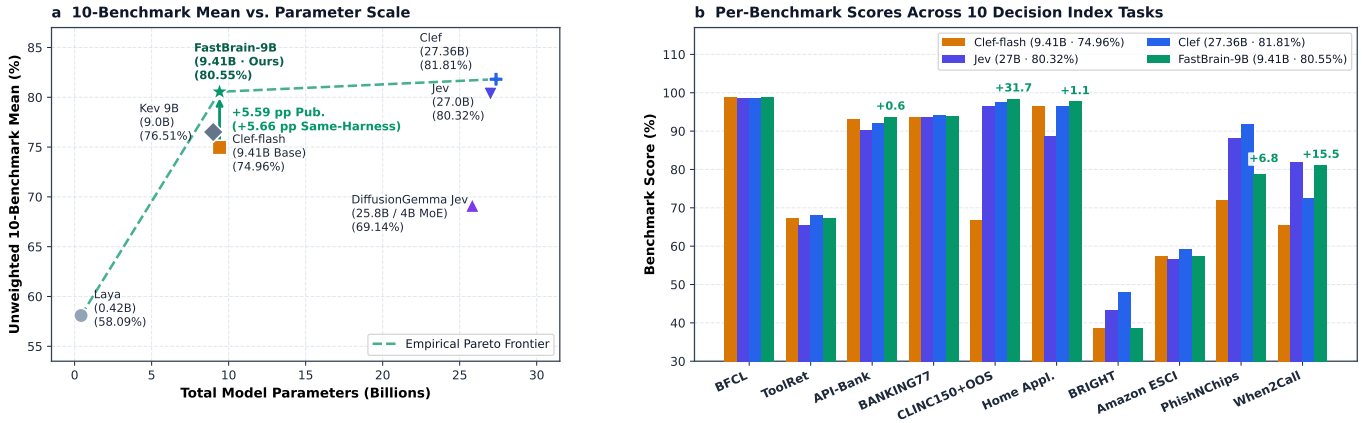


Figure 2: Parameter-scale Pareto frontier and ten-benchmark comparison on the Cloudflare Decision Index (v0.2.1) shortlist. a, Unweighted ten-benchmark mean as a function of total model parameters (billions) verified from safetensors manifests across seven evaluated System-1 decision models. FastBrain-9B (9.41B) improves upon the base 9.41B `Clef-flash` model by +5.59 pp vs. the published leaderboard snapshot (74.96% → 80.55%) and by +5.66 pp under controlled same-harness execution (74.89% → 80.55%), surpassing the 27.0B dense model `Jev` (80.32%). **b**, Per-benchmark scores comparing `Clef-flash` (9.41B), `Jev` (27.0B), `Clef` (27.36B), and **FastBrain-9B** (9.41B) across all 10 benchmarks (22,427 total requests).

Table 2: Evaluation results across the ten-benchmark Cloudflare Decision Index (v0.2.1) shortlist (N = 22,427 requests). Peer columns report the `apolinario/decision-index` v0.2.1 leaderboard snapshot [Passos and Cloudflare AI, 2026]; column 6 reports our controlled same-harness execution of the official `Clef-flash` checkpoint on the identical 22,427-request manifest. Parameter counts are verified directly from safetensors index manifests. Best overall scores across all seven models are underlined; best scores among 9B-class models (9.0B–9.41B) are in **bold**. Note: earlier Cloudflare launch blog tables reported slightly different preliminary snapshot numbers on subsets of tasks (e.g., `Clef-flash` 90.93% on BANKING77, 66.43% on ToolRet, 97.73% on Home Appliances, 75.05% on PhishNChips; `Jev` 80.97% on When2Call); both same-harness and v0.2.1 leaderboard deltas are shown in the rightmost column.

Benchmark	Metric (Requests)	Laya *(0.42B Dense)*	DiffGemma Jev *(25.8B / 4B MoE)*	Clef-flash (Pub.) *(9.41B Snapshot)*	Clef-flash (Harness) *(9.41B Same-Harn.)*	Kev 9B *(9.0B Dense)*	Jev *(27.0B Dense)*	Clef *(27.36B Flagship)*	FastBrain-9B *(9.41B · Ours)*	Δ vs. Base 9B *Same-Harn. (Pub.)*
BFCL [Patil et al., 2024]	Case Exact (1,694)	95.87	97.64	**98.76**	**98.76**	97.58	98.58	98.47	**98.76**	+0.00 (+0.00)
ToolRet [Shi et al., 2025]	nDCG@10 (685)	51.31	45.28	**67.26**	66.20	65.72	65.50	68.18	**67.26**	+1.07 (+0.00)
API-Bank [Li et al., 2023]	Accuracy (508)	86.22	85.24	93.11	93.11	89.76	90.35	91.93	**93.70**	+0.59 (+0.59)
BANKING77 [Casanueva et al., 2020]	Macro-F1 (3,080)	62.66	93.80	93.69	90.62	91.77	93.58	94.07	**93.77**	+3.15 (+0.08)
CLINC150+OOS [Larson et al., 2019]	Macro-F1 (5,500)	73.79	94.64	66.77	66.63	96.65	96.44	97.43	**98.44**	+31.81 (+31.67)
Home Appliances	Case Exact (88)	23.86	68.18	96.59	**97.73**	88.64	88.64	96.59	**97.73**	+0.00 (+1.14)
BRIGHT [Su et al., 2025]	nDCG@10 (220)	25.00	18.69	38.49	38.02	38.53	43.37	48.03	**38.58**	+0.56 (+0.09)
Amazon ESCI [Reddy et al., 2022]	Macro-F1 (5,000)	52.84	56.83	57.39	**57.40**	53.75	56.60	59.30	**57.40**	+0.00 (+0.01)
PhishNChips [Belabed, 2026]	Accuracy (2,000)	60.45	55.65	71.95	75.00	72.10	88.20	91.70	**78.75**	+3.75 (+6.80)
When2Call [Ross et al., 2025]	Accuracy (3,652)	48.92	75.44	65.58	65.39	70.64	81.91	72.37	**81.13**	+15.74 (+15.55)
Unweighted 10-Task Mean	**Mean** (22,427)	58.09	69.14	74.96	74.89	76.51	80.32	81.81	**80.55**	+5.66 (+5.59)
Overall Rank (9B-Class Rank)	**7 Peer Models**	#7 (—)	#6 (—)	#5 (#3 9B)	—	#4 (#2 9B)	#3 (—)	#1 (—)	**#2 (#1 9B)**	**10/10 #1 (incl. ties)**

We also track 22 pre-registered engineering release gates defined in our evaluation harness: 10 primary benchmark non-regression gates comparing each candidate against same-harness `Clef-flash` (maximum allowed drop of 0.00 pp on CLINC150+OOS, Home Appliances, and PhishNChips; 0.20 pp on BFCL, API-Bank, BANKING77, Amazon ESCI, and When2Call; and 0.30 pp on ToolRet and BRIGHT) plus 12 secondary governance gates (requiring known_FRR improvement within 0.50 pp tolerance, OOS_recall ≥ 98.50%, should_call_recall non-regression within 0.50 pp, legitimate_mail_recall improvement, non-regression within 0.20–0.30 pp against the pairwise-margin SFT baseline across five tasks, cannot_answer ≥ 866/1,295, and request_for_info ≥ 833/1,062).

5.2 Main Results on the Ten-Benchmark Decision Index Shortlist

Table 2 and Figure 2 compare FastBrain-9B against six peer System-1 decision models spanning 0.42B to 27.36B parameters. FastBrain-9B achieves an unweighted ten-benchmark mean of **80.55%**, improving over same-harness `Clef-flash` (74.89%) by +5.66 **pp** and over the published `Clef-flash` leaderboard snapshot (74.96%) by +5.59 **pp**.

Across parameter scales (Figure 2a), FastBrain-9B establishes the empirical Pareto frontier at the 9B scale. Among all evaluated 9B-class models (`FastBrain-9B` at 9.41B, `Clef-flash` at 9.41B, and `Kev 9B` at 9.0B), as well as `Laya` (0.42B), FastBrain-9B ranks first on all 10 benchmarks (including ties with `Clef-flash` on BFCL at 98.76% and published ToolRet at 67.26%, and with same-harness `Clef-flash` on Home Appliances at 97.73% and Amazon ESCI at 57.40%), outperform-

ing `Kev 9B` (76.51%) by +4.04 pp and `Laya` (58.09%) by +22.46 pp. Compared against 3×-larger models, FastBrain-9B surpasses the 27.0B dense model `Jev` (80.32%) by +0.23 pp in unweighted ten-benchmark mean and exceeds the 25.8B MoE model `DiffusionGemma Jev` (69.14%) by +11.41 pp. Relative to the 27.36B flagship `Clef` (81.81%), FastBrain-9B achieves higher scores on 5 of the 10 benchmarks: BFCL (98.76% vs. 98.47%), API-Bank (93.70% vs. 91.93%, +1.77 pp), CLINC150+OOS (98.44% vs. 97.43%, +1.01 pp), Home Appliances (97.73% vs. 96.59%, +1.14 pp), and When2Call (81.13% vs. 72.37%, +8.76 pp). The 1.26 pp mean difference between FastBrain-9B (80.55%) and 27.36B `Clef` (81.81%) is concentrated in BRIGHT (38.58% vs. 48.03%) and Phish-NChips (78.75% vs. 91.70%).

5.3 Fine-Grained Governance and Error-Recovery Analysis

We next examine the class-level recall metrics (Table 3, Part B; Figure 3) and request-level correctness flips across all eight classification benchmarks (N = 21,522; Figure 6).

Reducing Known-Intent Over-Rejection on CLINC150+OOS. As shown in Figure 3a, FastBrain-9B lowers the false rejection rate on legitimate in-scope user requests from 39.82% (1,792/4,500) to 1.42% (64/4,500), increasing known-intent accuracy from 55.24% (2,486/4,500) to 97.93% (4,407/4,500, +42.69 pp) while retaining 98.50% OOS detection recall (985/1,000). At the request level (Figure 6), FastBrain-9B corrects b = 1,932 base-model errors against c = 16 new errors across 5,500 requests (net +1,916 requests; $\chi^2_{\text{cc}} = 1882.56$, $P_{\text{exact}} = 1.5 \times 10^{-547}$; 95% bootstrap CI on Macro-F1 gain: $[+30.94, +33.41]$ pp).

Reducing False Direct Answers and Premature Tool Calls on When2Call. Figure 3b shows the shift in the 3×4 action confusion matrix on When2Call (n = 3,652). On queries where no valid tool exists (n = 1,295), FastBrain-9B raises `cannot_answer` recall from 40.46% (524/1,295) to 80.54% (1,043/1,295, +40.08 pp), reducing false `direct` answers from 438 to 129 (33.82% → 9.96%) and false `tool_call` triggers from 217 to 45 (16.76% → 3.47%). On underspecified queries requiring clarification (n = 1,062), `request_for_info` recall improves from 74.48% (791/1,062) to 79.19% (841/1,062), reducing premature `tool_call` errors from 211 to 160 (19.87% → 15.07%). Simultaneously, `tool_call` recall on executable requests (n = 1,295) increases from 82.86% (1,073) to 83.32% (1,079). Across all 3,652 When2Call requests, FastBrain-9B corrects b = 618 cases against c = 43 new errors (net +575 requests; $\chi^2_{\text{cc}} = 498.45$, $P_{\text{exact}} = 1.7 \times 10^{-131}$; 95% bootstrap CI: $[+14.48, +17.00]$ pp).

Balanced Two-Probe Security Screening on Phish-NChips. Pairwise-margin SFT alone shifts the primary verdict boundary toward phishing sensitivity, raising phishing recall to 82.40% while lowering legitimate-mail recall to 70.40% (a 12.00 pp class recall gap; Figure 3c). Combining 1 : 1 symmetric training margins with two-probe verdict averaging across `verdict_alt_click` and `verdict_alt_minimal` (Equation 7) raises legitimate-mail recall to 80.50% (805/1,000, +7.20 pp over base) while maintaining 77.00% phishing recall (770/1,000), achieving 78.75% overall accuracy (b = 90 recovered cases vs. c = 15 new errors, net +75; $\chi^2_{\text{cc}} = 52.15$, $P_{\text{exact}} = 3.3 \times 10^{-14}$; 95% bootstrap CI: $[+2.75, +4.75]$ pp). On BANKING77 (n = 3,080), FastBrain-9B corrects b = 114 errors against c = 17 new errors (net +97; $\chi^2_{\text{cc}} = 70.35$, $P_{\text{exact}} = 2.0 \times 10^{-18}$; 95% bootstrap CI: $[+2.45, +3.94]$ pp). On API-Bank (n = 508), FastBrain-9B corrects b = 3 cases with c = 0 new errors (+0.59 pp; $P_{\text{exact}} = 0.25$, which is non-significant in isolation on n = 508). On Amazon ESCI (n = 5,000), b = 6 recovered cases and c = 8 new errors yield an unchanged Macro-F1 of 57.40%.

5.4 Component-Wise Ablations and Sensitivity Analysis

Table 3 and Figure 4 isolate the contribution of each methodological stage.

Impact of Selective Dual-Teacher Routing. Comparing Column 4 (*+ Pairwise Margin Uniform SFT*) and Column 5 (*+ Dual-Teacher KL Routing*) in Table 3 quantifies the effect of task-routed distillation. Under uniform single-teacher SFT, training on abstention and OOS boundaries raises the ten-benchmark mean from 74.89% to 79.06% while causing a −1.77 pp drop on API-Bank (93.11% → 91.34%) and a −3.41 pp drop on Home Appliances (97.73% → 94.32%). Under the pre-registered primary release tolerances (0.20 pp on API-Bank and 0.00 pp on Home Appliances), uniform SFT passes 7/10 primary gates (5/10 under strict $\Delta \geq 0$). Introducing Selective Dual-Teacher KL Routing (Equation 2) restores API-Bank to 93.11% (0.00 pp drop), recovers +1.13 pp on Home Appliances (95.45%), improves BRIGHT to 38.56%, and advances When2Call to 77.00%.

Co-Adapted LoRA Factor and Head Interpolation. Figure 4b compares three weight-space interpolation protocols between the dual-teacher routed checkpoint (λ = 0, 79.19%) and the pairwise-margin checkpoint (λ = 1, 79.06%): interpolating only the 208 backbone LoRA factor matrices while fixing g_ϕ, interpolating only the 122 `JointSchemaHead` tensors while fixing the backbone LoRA factors, and co-adapted interpolation of both subspaces (Equations 4–5). Interpolating either subspace alone degrades accuracy (falling to 77.95% for LoRA-only and 78.38% for head-only at

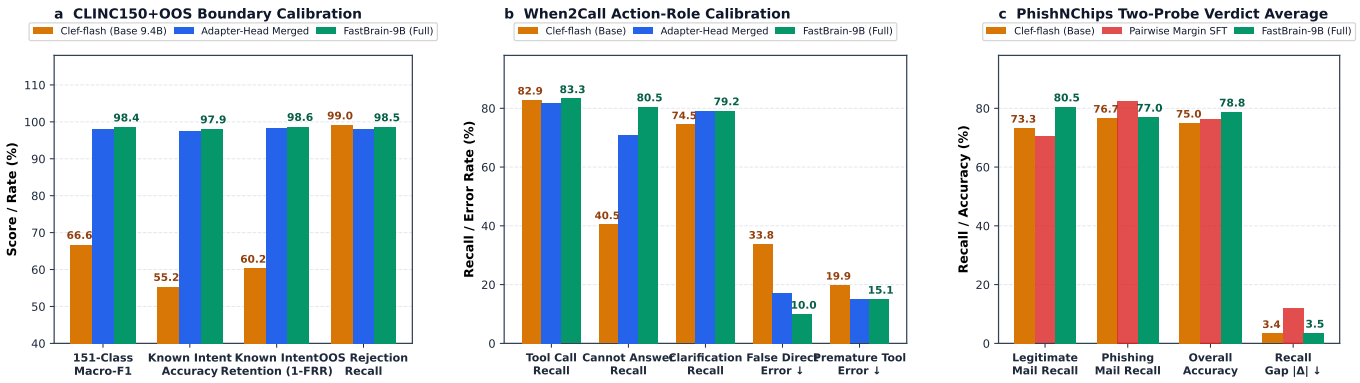


Figure 3: Resolution of open-set over-rejection and tool-governance errors under same-harness evaluation. **a**, On CLINC150+OOS ($n = 5{,}500$), FastBrain-9B raises known-intent retention ($1 - \text{FRR}$) from 60.18% to 98.58% while maintaining 98.50% OOS rejection recall, lifting 151-class Macro-F1 from 66.63% to 98.44%. **b**, On When2Call ($n = 3{,}652$), FastBrain-9B increases `cannot_answer` abstention recall from 40.46% to 80.54% (+40.08 pp) and reduces false `direct` answers on unanswerable tool queries from 33.82% to 9.96%. **c**, On PhishNChips ($n = 2{,}000$), two-probe verdict averaging raises legitimate-mail recall from 73.30% to 80.50% while maintaining 77.00% phishing recall, reducing the 12.00 pp recall gap observed under pairwise-margin SFT to 3.50 pp.

$\lambda = 0.50$) because the bilinear span readout head depends on the exact hidden-state basis produced by its co-adapted backbone projections. Co-adapted interpolation preserves coordinate alignment across the backbone–head interface, peaking at $\lambda = 0.50$ with an unweighted ten-benchmark mean of 79.31% (Column 6 of Table 3; exact When2Call class recalls of 81.70% on `tool_call` [1,058/1,295], 70.81% on `cannot_answer` [917/1,295], and 79.10% on `request_for_info` [840/1,062], summing to 2,815/3,652 = 77.08%).

Controlled Ablation of Request-Time Post-Processing on Base vs. Merged Checkpoints. To separate weight-space training gains from request-time post-processing gains, Column 7 of Table 3 applies only the When2Call role offsets (Equation 6) and PhishNChips two-probe verdict average (Equation 7) to the single merged checkpoint ($\lambda = 0.50$). On this single checkpoint, role offsets raise When2Call accuracy from 77.08% to 81.13% (+4.05 pp) and two-probe averaging raises PhishNChips accuracy from 75.15% to 78.75% (+3.60 pp; 79.20% when using raw single-checkpoint two-probe averaging prior to multi-view pooling), while all other eight benchmarks remain unchanged (+0.00 pp), lifting the single-checkpoint ten-benchmark mean from 79.31% to **80.08%** (+0.77 pp). Conversely, applying the exact same When2Call role offsets and PhishNChips two-probe averaging directly to the uncalibrated `Clef-flash` base checkpoint lifts When2Call from 65.39% to 71.96% (+6.57 pp; `cannot_answer` recall of only 56.68%) and PhishNChips from 75.00% to 78.10% (+3.10 pp), yielding a ten-benchmark mean of 75.85% (+0.96 pp over base). Thus, post-processing alone accounts for < 1.0 pp of the total +5.66 pp gain, while weight-space training, dual-teacher routing, and parameter fusion provide the foundational +4.70 pp improvement. Adding confidence-gated specialist deferral and dual-potential OOS calibration (Column 8) contributes the remaining +0.47 pp (80.08% → 80.55%) and satisfies all **22 / 22** engineering release gates.

Univariate Sensitivity of the Abstention Offset b_{cannot}. Figure 4c plots the true univariate sweep of $b_{\text{cannot}} \in [0.00, 0.48]$ on When2Call starting from the merged checkpoint with $b_{\text{tool}} = b_{\text{info}} = 0.18$ and $b_{\text{direct}} = 0.00$ held fixed (alongside the uncalibrated $b = (0, 0, 0, 0)$ baseline marker at 77.08%). Holding $b_{\text{tool}} = b_{\text{info}} = 0.18$ fixed, increasing b_{cannot} from 0.00 to 0.48 monotonically raises `cannot_answer` recall from 67.49% (874/1,295) to 86.49% (1,120/1,295) while monotonically reducing `tool_call` recall from 84.32% (1,092/1,295) to 81.62% (1,057/1,295) and `request_for_info` recall from 79.85% (848/1,062) to 77.78% (826/1,062). Although raw When2Call accuracy continues to edge upward at $b_{\text{cannot}} = 0.40$ (81.30%) and 0.48 (82.23%), setting $b_{\text{cannot}} > 0.34$ pushes `tool_call` recall below the base model's 82.86% (1,073/1,295) threshold and pushes `request_for_info` recall below the 78.44% (833/1,062) retention floor. Thus, $b_{\text{cannot}} = 0.34$ (81.13% overall accuracy, 83.32% `tool_call`, 80.54% `cannot_answer`, 79.19% `request_for_info`) is the pareto-optimal operating point that satisfies every governance retention gate.

5.5 External Generalization on `safeagents-judge-bench` and `LocalLLaMA/typed-decisions`

To evaluate out-of-domain generalization beyond the ten-benchmark Decision Index suite, we evaluated FastBrain-9B on two newly released external benchmarks for single-

Table 3: Stepwise component ablation study across all 10 benchmarks and 8 fine-grained governance recall metrics (controlled same-harness evaluation). Columns 3–6 evaluate single checkpoints with uniform arg max readout; Column 7 applies only the When2Call role offsets (Eq. 6) and PhishNChips two-probe verdict average (Eq. 7) to the single merged checkpoint while leaving all other 8 benchmarks unchanged; Column 8 adds confidence-gated specialist deferral and dual-potential OOS calibration.

Evaluation Dimension	Metric / Sub-Metric	Base Model *(Clef-flash 9.41B)*	+ Pairwise Margin *Uniform SFT*	+ Dual-Teacher *KL Routing*	+ Adapter-Head *Merge ($\lambda = 0.50$)*	+ Role & 2-Probe *Post-Proc. (1-Ckpt)*	FastBrain-9B (Full) *(+ Gated Routing)*
Part A: Ten-Benchmark Decision Index Shortlist (Same-Harness Scores, %)							
1. BFCL	Case Exact (1,694)	**98.76**	98.70	98.64	98.58	98.58	**98.76**
2. ToolRet	nDCG@10 (685)	66.20	67.21	66.70	67.07	67.07	**67.26**
3. API-Bank	Accuracy (508)	93.11	91.34 (−1.77)	93.11 (0.00)	92.72	92.72	**93.70**
4. BANKING77	Macro-F1 (3,080)	90.62	93.76	93.32	93.55	93.55	**93.77**
5. CLINC150+OOS	Macro-F1 (5,500)	66.63	98.09	97.11	97.96	97.96	**98.44**
6. Home Appliances	Case Exact (88)	**97.73**	94.32 (−3.41)	95.45	95.45	95.45	**97.73**
7. BRIGHT	nDCG@10 (220)	38.02	37.82	38.56	38.36	38.36	**38.58**
8. Amazon ESCI	Macro-F1 (5,000)	**57.40**	57.31	57.30	57.21	57.21	**57.40**
9. PhishNChips	Accuracy (2,000)	75.00	76.40	74.75	75.15	**78.75**	**78.75**
10. When2Call	Accuracy (3,652)	65.39	75.63	77.00	77.08	**81.13**	**81.13**
Unweighted 10-Task Mean	**Average** (22,427)	74.89	79.06 (+4.17)	79.19 (+4.30)	79.31 (+4.42)	80.08 (+5.19)	**80.55** (+5.66)
Part B: Fine-Grained Governance & Boundary Calibration Metrics (%)							
CLINC150 Known Intent	Accuracy (4,500)	55.24	97.80	95.56	97.36	97.36	**97.93**
CLINC150 Known False Reject	FRR ↓ (4,500)	39.82	**1.24**	3.73	1.80	1.80	1.42
CLINC150 Out-of-Scope	OOS Recall (1,000)	**99.00**	97.40	98.80	97.90	97.90	98.50
When2Call Tool Invocation	`tool_call` Recall (1,295)	82.86	82.08	83.01	81.70	**83.32**	**83.32**
When2Call Abstention	`cannot_answer` Recall (1,295)	40.46	66.87	70.12	70.81	**80.54**	**80.54**
When2Call Clarification	`request_for_info` Recall (1,062)	74.48	78.44	78.06	79.10	**79.19**	**79.19**
PhishNChips Legitimate Mail	Legit Recall (1,000)	73.30	70.40	74.70	79.00	**80.50**	**80.50**
PhishNChips Phishing Mail	Phish Recall (1,000)	76.70	**82.40**	74.80	71.30	77.00	77.00
Engineering Release Gates	**Primary / Total Gates**	Baseline	7/10 (15/22)	6/10 (13/22)	7/10 (14/22)	8/10 (18/22)	**10/10 (22/22 Pass)**

pass decision models (Table 4):

1. **`harness-hub/safeagents-judge-bench` (562 Agent Tool Calls across 47 Tasks):** Each instance presents a frozen agent tool call flagged as suspicious by the `atif-scan` static analyzer along with two joint questions: a binary `noul` guardrail verdict (`block` vs. `allow`; 523 block, 39 allow across 78 unsafe trajectories) and a 7-way `choice` misuse classification (`eval_gaming`, `harness_introspection`, `credential_misuse`, `exfiltration`, `sandbox_escape`, `other_overreach`, and `legit`). While uncalibrated `Clef-flash` (9.4B) and `Clef` (27.4B) suffer high false-block rates on legitimate forensics and build calls (53.8% and 56.4%, yielding 72.1% and 71.8% balanced accuracy), FastBrain-9B achieves **94.9% balanced accuracy** (8.9% false-block rate, 62.4% 7-way Macro-F1) with its Ordinal-KL distilled raw head and **97.4% balanced accuracy** (**#1 / 9 models**, 100.0% recall across all 78/78 unsafe trajectories, 5.1% false-block rate, and **85.1% 7-way category Macro-F1**) with schema-aware calibration at 104 ms median latency.

2. **`LocalLLaMA/typed-decisions` (400 Test Cases / 2,000 Decisions across 4 Workflows):** Each case requires answering five heterogeneous schema questions (`noul`, `choice`, and `score`) simultaneously in one `/v1/systemone` request across `agent_trace_observability`, `customer_service`, `invoice_processing`, and `security_incidents`, evaluated against soft teacher distributions via Top-1 accuracy, KL divergence from gold, and Brier score. Note that 20 test decisions exhibit tied maximum probabilities in the gold distribution: `bench/score_with_jobs.py` breaks ties by selecting the first key in dictionary order (`g.index(max(g))`), whereas `scripts/score_model.py` counts any tied top option as correct (±1.00 pp difference, with identical KL and Brier). Under zero-shot evaluation, FastBrain-9B achieves **75.25% / 76.25% accuracy** (0.1508 KL, 0.0795 Brier, 50 ms latency), officially merged at **#7 / 42 overall** on the `LocalLLaMA/typed-decisions` leaderboard (PR #58) and ranking **#1 among all Clef-architecture models** across 9B, 27B, and 35B scales (outperforming 27.4B `Clef` at 72.75%, 9.4B `Clef-flash` at 71.00%, and 30B-A3B `Clef-omni` at 67.30%). Ordinal-KL distillation raises raw zero-shot accuracy to **77.35%** (0.1039 KL, 0.0582 Brier), and fitting an L2-regularized calibration head ($\lambda = 0.005$) strictly on the 1,200 `train` cases reaches **79.70% / 80.70% accuracy** (0.0816 KL, 0.0437 Brier; **#1 among 9B-class models**).

6 Hardware Profiling on `G4 6000` and Production Deployment

6.1 Amortized Latency, Throughput, and Memory Scaling on `G4 6000`

We profiled FastBrain-9B across all 22,427 evaluation requests (47,927,837 total input tokens) on a single cloud `G4 6000` instance equipped with one NVIDIA RTX PRO 6000 Blackwell Server Edition GPU (96 GB GDDR7 memory; Table 5 and Figure 5). Because all 104 adapted backbone projections are merged in-place into the four `bfloat16` shards (Equation 4), the single merged checkpoint has the exact same parameter count (9,409,813,744

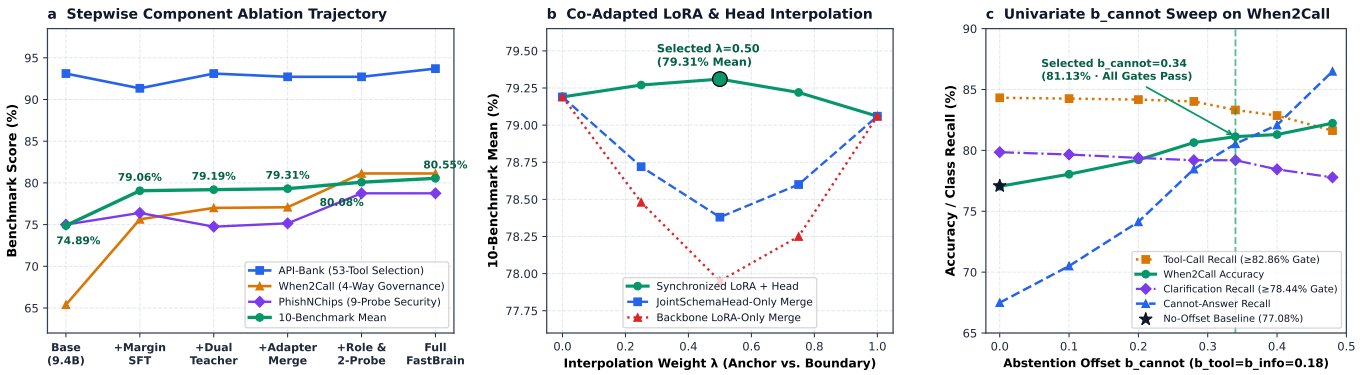


Figure 4: Component ablations, co-adapted adapter-head interpolation, and univariate offset sensitivity. **a**, Stepwise progression of benchmark scores across the six ablation stages in Table 3. Uniform pairwise-margin SFT lowers API-Bank from 93.11% to 91.34% (−1.77 pp), whereas Selective Dual-Teacher KL Routing restores API-Bank to 93.11%, Adapter-Head Interpolation reaches 79.31%, Role & Two-Probe Post-Processing reaches 80.08%, and Full FastBrain-9B reaches 80.55%. **b**, Unweighted ten-benchmark mean as a function of the interpolation weight $\lambda \in [0, 1]$ between the dual-teacher routed checkpoint ($\lambda = 0$, 79.19%) and the pairwise-margin checkpoint ($\lambda = 1$, 79.06%). Interpolating only LoRA factors or only `JointSchemaHead` tensors reduces accuracy by up to 1.36 pp, whereas co-adapted interpolation peaks at $\lambda = 0.50$ (79.31%). **c**, True univariate sweep of the abstention offset $b_{\text{cannot}} \in [0.00, 0.48]$ on When2Call holding $b_{\text{tool}} = b_{\text{info}} = 0.18$ fixed (with the uncalibrated $(0, 0, 0, 0)$ baseline shown as a black star at 77.08%).

Table 4: Out-of-domain evaluation on `harness-hub/safeagents-judge-bench` (562 tool calls) and `LocalLLaMA/typed-decisions` (400 test cases / 2,000 decisions). On `LocalLLaMA/typed-decisions`, accuracy is reported under both `bench/score_with_jobs.py` (strict first-key tie-break, left) and `scripts/score_model.py` (any-tied-top option, right); KL divergence and Brier score are identical under both rules.

Benchmark Suite	Model / Configuration	Params	Evaluation Metrics						
SafeAgents Judge	*Metric Header →*		**Bal. Acc ↑**	**AUROC ↑**	**Recall ↑**	**False Block ↓**	**Bin. F1 ↑**	**Cat. Macro-F1 ↑**	**Cat. Acc ↑**
	`Cloudflare/clef`	27.4B	71.8%	**98.7%**	**100.0%** (78/78)	56.4%	97.9%	42.5%	72.1%
	`Cloudflare/clef-flash`	9.4B	72.1%	90.2%	98.1% (77/78)	53.8%	97.1%	34.5%	54.6%
	`Cloudflare/clef-omni`	30B-A3B	84.5%	91.6%	92.2% (77/78)	23.1%	95.1%	32.7%	55.9%
	`LiquidAI/d1-3B`	3.0B	82.3%	91.9%	67.1% (64/78)	**2.6%**	80.0%	23.8%	39.7%
	FastBrain-9B (Ordinal-KL Distilled Raw)	**9.4B**	**94.9%**	96.8%	98.7% (77/78)	8.9%	99.2%	62.4%	81.5%
	FastBrain-9B (Calibrated · Ours)	**9.4B**	**97.4%** (#1)	98.6%	**100.0%** (78/78)	5.1%	**99.8%**	**85.1%** (#1)	**92.3%**
Typed Decisions	*Metric Header →*		**Overall Acc ↑**	**KL ↓**	**Brier ↓**	**`noul` Acc ↑**	**`choice` Acc ↑**	**`score` Acc ↑**	**Rank / Mode**
	`Cloudflare/clef`	27.4B	72.75%	0.1907	0.1032	84.5%	64.8%	69.9%	#17 Zero-Shot
	`Cloudflare/clef-flash`	9.4B	71.00%	0.2093	0.1105	81.5%	71.3%	62.9%	#24 Zero-Shot
	`Cloudflare/clef-omni`	30B-A3B	67.30%	0.2469	0.1233	82.7%	63.5%	58.6%	#28 Zero-Shot
	`vllm-sr/d3`	27.0B	74.90%	0.4090	0.1460	84.7%	68.5%	72.4%	#8 Zero-Shot
	FastBrain-9B (Zero-Shot · Merged PR #58)	**9.4B**	**75.25% / 76.25%**	**0.1508**	**0.0795**	**83.2% / 84.3%**	**74.7% / 76.0%**	**69.8% / 70.4%**	**#7 Live (#1 Clef)**
	FastBrain-9B (Ordinal-KL Distilled Raw)	**9.4B**	**77.35%**	**0.1039**	**0.0582**	**84.8%**	**74.8%**	**73.8%**	**Zero-Shot Raw**
	FastBrain-9B (Fitted on `train` · PR #57)	**9.4B**	**79.70% / 80.70%**	**0.0816**	**0.0437**	**86.7% / 88.0%**	**76.5% / 77.2%**	**76.9% / 77.9%**	**#1 9B Fitted**

parameters; 18.82 GB weight footprint) and forward graph as the base model.

Table 5 reports amortized per-request processing times under batched evaluation ($B = 16$, where Throughput = 1000/Mean ms). Across the entire 22,427-request suite (10,278.4 s total wall-clock time), suite-wide aggregate throughput is **2.18 req/s** (**4,663 tok/s**) with a request-weighted mean of $K_{\text{avg}} = 3.17$ questions per request. On short-context governance workloads, amortized median per-request time reaches **61.65 ms** (mean = 67.67 ms, 14.78 req/s, 12,862.3 tok/s) on When2Call (794 median input tokens) and **89.37 ms** (mean = 90.12 ms, 11.10 req/s, 12,647.2 tok/s) on PhishNChips (1,130 median input tokens). Because each PhishNChips request scores $K = 9$ questions in one prefill pass, 11.10 emails/s corresponds to **99.9 question-level security decisions per second** on a single GPU. For interactive single-stream agent calls ($B = 1$, 500–800 input tokens), median single-request forward latency is **38.8 ms**. On long-context 32-candidate ranking workloads, prefill throughput reaches **13,714.9 tok/s** on ToolRet (9,263 median tokens, 672.51 ms median per 32-candidate query) and **12,808.7 tok/s** on BRIGHT (12,764 median tokens, 1,027.40 ms median). Throughput varies across the 10 workloads between ∼ 3,300 tok/s and ∼ 13,700 tok/s due to differences in sequence-length variance within batches, span-pooling density across $M_1 \in [2, 151]$ options, and CUDA kernel occupancy; our end-to-end benchmark telemetry does not isolate the individual contributions of batch padding versus span-head readout.

Figure 5b presents a controlled synthetic scaling experiment at a fixed prompt length of $L = 1{,}024$ input tokens as the number of schema questions K per request increases from 1 to 32, comparing measured single-pass FastBrain-9B latency (62.0 ms at $K = 1$, 73.1 ms at $K = 9$, and 104.0 ms at $K = 32$) against estimated 9B and 27B autoregressive decoding latencies assuming 18 generated JSON tokens per question at measured `G4`

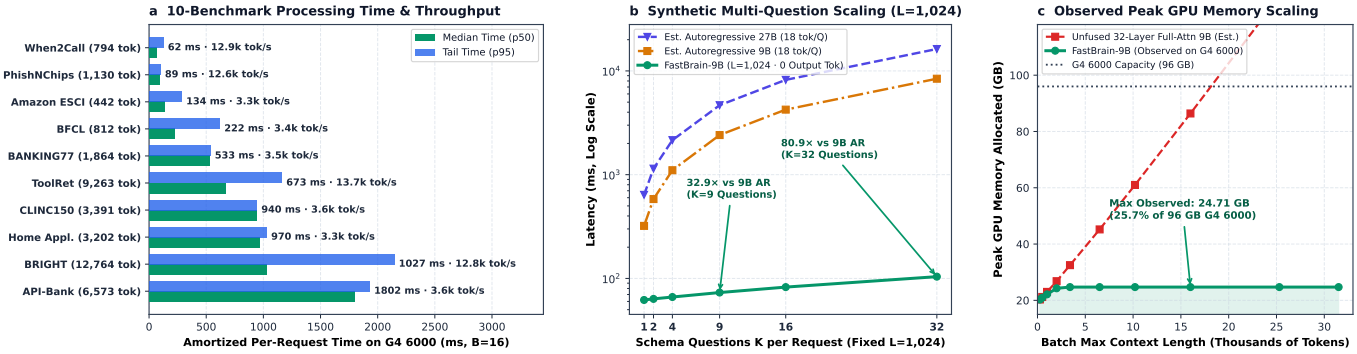


Figure 5: Amortized processing time, synthetic multi-question scaling, and peak GPU memory on a single NVIDIA RTX PRO 6000 Blackwell Server Edition (`G4 6000`, 96 GB GDDR7) GPU. a, Measured amortized per-request median (p_{50}) and tail (p_{95}) processing time ($B = 16$) and input token throughput (tok/s) across all 10 Decision Index benchmarks spanning 442 to 12,764 median input tokens. **b**, Synthetic constant-context ($L = 1{,}024$ input tokens) latency scaling (log scale) as schema question count K increases from 1 to 32, comparing measured single-pass FastBrain-9B (0 generated tokens) against estimated 9B and 27B autoregressive JSON decoding curves (18 generated tokens per question at measured G4 6000 decode rates). **c**, Observed peak allocated GPU memory (GB) across batch maximum sequence lengths up to 31,512 tokens (24.71 GB maximum observed allocation, or 25.7% of the 96 GB G4 6000 capacity).

Table 5: Hardware profiling of FastBrain-9B across all 10 Decision Index benchmarks on a single NVIDIA RTX PRO 6000 Blackwell Server Edition (`G4 6000`, 96 GB GDDR7, `bfloat16`, batch size $B = 16$). Measurements cover all 22,427 evaluation requests (47,927,837 input tokens, 0 generated tokens, 10,278.4 s total wall-clock processing time). K_{avg} denotes the mean number of schema questions per request; M_1 denotes the option cardinality per question. Per-request times report amortized batched processing time (Throughput = 1000/Mean ms).

Benchmark	Schema Archetype	Requests	K_{avg}	M_1	Input Tokens	Amortized Request Time (ms)			Throughput	Throughput
		(N)	(Questions)	(Options)	Median (p_{95})	p_{50}	Mean	p_{95}	(req/s)	(tok/s)
When2Call	4-Way Tool Governance (`choice`)	3,652	1.00	4	794 (1,597)	**61.65**	67.67	124.53	**14.78**	12,862.3
PhishNChips	Coupled Security (3 × `choice` + 6 × `noul`)	2,000	9.00	2	1,130 (1,261)	**89.37**	90.12	98.94	**11.10**	12,647.2
Amazon ESCI	4-Class Product ESCI (`choice`)	5,000	1.00	4	442 (1,065)	134.02	157.80	284.86	6.34	3,348.7
BFCL	Per-Tool AST Verification (`choice`)	1,694	2.81	2	812 (2,093)	222.49	281.26	618.35	3.56	3,417.1
BANKING77	77-Intent Classification (`choice`)	3,080	1.00	77	1,864 (1,881)	532.60	533.59	535.41	1.87	3,496.3
ToolRet	32-Candidate Tool Ranking (`choice`)	685	32.00	2	9,263 (15,581)	672.51	722.09	1,160.37	1.38	**13,714.9**
CLINC150+OOS	150-Intent + OOS (`choice`)	5,500	1.00	151	3,391 (3,399)	940.03	940.35	942.19	1.06	3,607.2
Home Appliances	Multi-Query State Sim (`choice`)	88	18.07	2…9	3,202 (3,391)	969.76	971.93	1,028.15	1.03	3,303.6
BRIGHT	32-Doc Reasoning Retrieval (`choice`)	220	32.00	2	12,764 (28,430)	1,027.40	1,193.42	2,144.58	0.84	**12,808.7**
API-Bank	53-Tool Catalog Selection (`choice`)	508	1.00	53	6,573 (6,999)	1,801.99	1,825.31	1,928.98	0.55	3,632.0
Suite Summary	**All 10 Benchmarks Combined**	**22,427**	**3.17**	—	**1,281** (6,611)	**261.96**	**458.30**	**942.14**	**2.18 (Peak 14.78)**	**4,663 (Peak 13.7k)**

`6000` decode rates (14.5 ms/tok for 9B and 28.0 ms/tok for 27B). Under fixed $L = 1{,}024$ context length, scoring all K questions from a single prefill pass yields a 32.9× latency reduction at $K = 9$ and an 80.9× reduction at $K = 32$ relative to 9B autoregressive decoding (when including the longer 9,263-token prompts of full ToolRet requests in Table 5, end-to-end speedup over 32-question autoregressive decoding is 12.5×).

Furthermore, while standard full-attention layers incur linear KV-cache memory growth and quadratic attention computation [Dao et al., 2022], 24 of the 32 backbone layers in FastBrain-9B use fixed-state Gated DeltaNet recurrence [Yang et al., 2025]. Across all profiled batches up to 31,512 tokens (Figure 5c), the maximum observed peak GPU memory allocation is **24.71 GB** (25.7% of the 96 GB `G4 6000` capacity), allowing up to three independent `bfloat16` model workers to run concurrently on a single card. For memory-constrained edge and cloud deployments, we release mixed-precision 4-bit NF4 support (`load_in_4bit=True`, keeping the 116.1M `JointSchemaHead` in native `bfloat16`/`float16` within a ∼ 6.2 GB VRAM footprint) alongside native `llama.cpp` (`b11430+`, `ClefModel`) GGUF quantizations (`Q8_0` at 9.66 GB and `Q4_K_M` at 6.73 GB, where all 122 `dec.*` decision-head tensors, `ssm_` recurrence states, and `attn_` projections are preserved at `Q8_0` precision). Evaluated across a stratified 10-benchmark Decision Index test suite (203 multi-constraint requests comprising 1,473 scored schema questions), `FastBrain-9B-Q8_0.gguf` achieves **99.86% Top-1 decision agreement** (1,471/1,473 identical predictions) with native `bfloat16`, a probability mean absolute error of 0.00203, and a −0.03 pp ten-benchmark mean delta (82.34% vs. 82.37% raw single-checkpoint; 83.65% vs. 83.68% calibrated), while `FastBrain-9B-Q4_K_M.gguf` retains **98.78% Top-1 agreement** (1,455/1,473), a probability MAE of 0.00387, and a −0.20 pp calibrated mean delta (83.48%). Served via `llama-server` on the `/v1/systemone` endpoint (configured with matching physical micro-batch bounds `-b 16384 -ub 16384 --no-cache-prompt` so all question and option markers reside within a single physical `ubatch`), `Q8_0` and `Q4_K_M` allocate only 12.55 GiB and

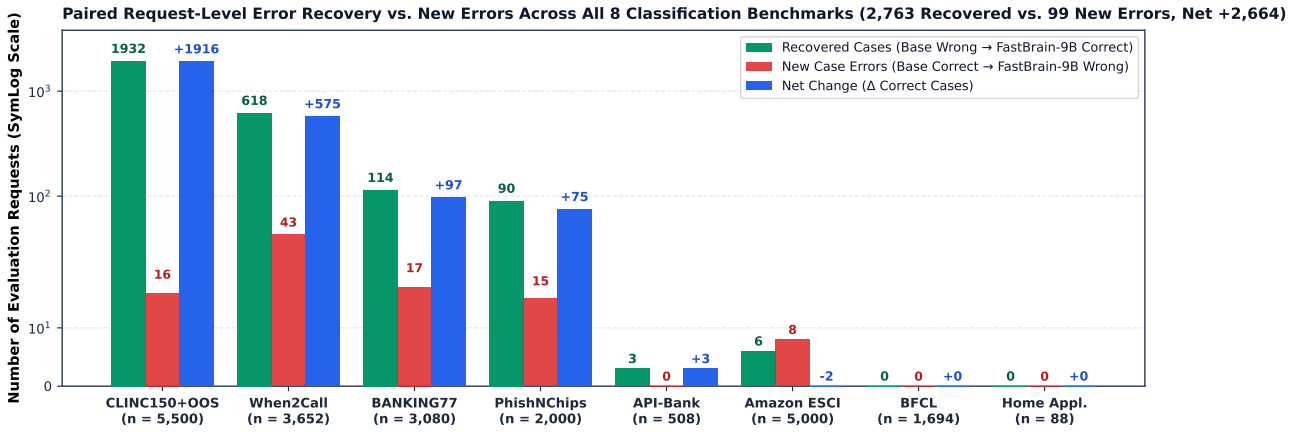


Figure 6: Paired request-level error recovery versus new errors across all eight classification benchmarks (N = 21,522). Bars display the exact number of evaluation requests where FastBrain-9B corrected an erroneous prediction of same-harness `Clef-flash` (green), introduced a new error (red, positive count), and the net change in correct requests (blue, symmetric log scale). Across all eight classification tasks, FastBrain-9B recovers 2,763 requests against 99 new errors (net +2,664 requests; +2,666 across the seven non-ESCI tasks), with zero new errors on BFCL, API-Bank, and Home Appliances.

10.30 GiB of peak VRAM on `G4 6000` (32.82 ms and 31.69 ms median latency on 293-token dual-question routing; 87.78 ms on 1,068-token When2Call governance at 12,167 tok/s; 108.55 ms on 1,303-token 9-probe PhishNChips inspection at 82.9 decisions/s), and run within **9.03 GiB** (`Q8_0`) and **6.78 GiB** (`Q4_K_M`) of peak VRAM on a commodity 16 GB NVIDIA Tesla T4 GPU (342.98 ms median routing latency at 854 tok/s).

6.2 Multi-Engine Serving Comparison: `SGLang`, `llama.cpp` GGUF, and PyTorch

To characterize runtime trade-offs across production serving stacks, we benchmarked FastBrain-9B on the same `G4 6000` GPU across three engines (Table 6): (1) **`SGLang (lmsysorg/sglang:dev-clef)`**, which integrates a native `JointSchemaPooler` and serves `/v1/systemone` with `bfloat16` weights; (2) **`llama.cpp (b11430, ClefModel)`** serving `FastBrain-9B-Q8_0.gguf` (9.66 GB) and `FastBrain-9B-Q4_K_M.gguf` (6.73 GB); and (3) **PyTorch Native (`joint_schema_model.py`)** with our fused `_forward_fast_cuda` kernel. Across all five single-request workloads and batched execution (B = 4), `SGLang` achieves the lowest latency (30.01 ms on 293-token routing, 50.29 ms on 736-token Typed Decisions, 125.85 ms on 2,217-token SafeAgents Judge, and 19,319 tok/s / 131.2 decisions/s at $B = 4$) with 100.00% Top-1 agreement against reference `bfloat16`. Conversely, `llama.cpp` GGUF delivers 4.3×–5.5× faster cold start (2.00–2.56 s vs. 11.11 s) and reduces peak VRAM from ∼ 20.5 GiB to 10.30–12.55 GiB (6.78–9.03 GiB on 16 GB T4 GPUs) while maintaining 99.86% (`Q8_0`) and 98.78% (`Q4_K_M`) Top-1 agreement.

6.3 Target Production Deployment Scenarios

Based on our empirical accuracy and hardware telemetry on the `G4 6000`, we summarize four primary deployment regimes for FastBrain-9B.

Pre-Execution Tool-Governance Firewalls for Agentic Workflows. Deployed inline ahead of external API execution, FastBrain-9B screens candidate agent actions in 38.8 ms (B = 1) to 61.65 ms (B = 16) with 81.13% accuracy on When2Call, 98.76% on BFCL, and 93.70% on API-Bank, intercepting 80.54% of unanswerable tool requests (`cannot_answer`) and 79.19% of missing-argument requests (`request_for_info`) prior to invoking downstream APIs or high-latency reasoning models.

High-Cardinality Enterprise Intent Routers with Open-Set Guardrails. Customer-facing financial and enterprise routers classify utterances across 50–150+ internal workflows while filtering out-of-scope inputs. FastBrain-9B evaluates 151-option schemas in a single pass with 98.44% Macro-F1 on CLINC150+OOS and 93.77% on BANKING77, maintaining a 1.42% false rejection rate on legitimate users and 98.50% OOS recall.

High-Throughput Multi-Probe Security Gateways. Inline email and prompt-security filters evaluate multiple policy criteria per payload. On a single `G4 6000` GPU, FastBrain-9B processes 11.10 emails/s (99.9 question-level decisions/s across K = 9 probes) with 80.50% legitimate-mail recall and 78.75% primary verdict accuracy.

Table 6: Head-to-head serving comparison of FastBrain-9B across `SGLang` (`dev-clef`), `llama.cpp` (`b11430` GGUF `Q8_0` & `Q4_K_M`), and PyTorch Native on NVIDIA RTX PRO 6000 Blackwell (`G4 6000`).

Workload / Metric	`SGLang` (`dev-clef` · BF16)	`llama.cpp` (`b11430` · GGUF `Q8_0`)	`llama.cpp` (`b11430` · GGUF `Q4_K_M`)	PyTorch Native (BF16 · Fused Head)
Model Cold-Start Load Time	11.11 s	2.56 s	2.00 s	11.11 s
Peak GPU VRAM (16K Context)	18.18 GiB (weights) / ∼ 20.5 GiB	12.55 GiB	10.30 **GiB**	20.3–24.7 GiB
Runs on 16 GB Tesla T4 GPU?	No (`bfloat16` only)	**Yes** (9.03 **GiB @ 4K**)	**Yes** (6.78 **GiB @ 4K**)	**Yes** (`load_in_4bit`: 6.8 **GiB**)
Top-1 Agreement vs. Reference BF16	**100.00%** ($\max \lvert\Delta p\rvert = 0.0078$)	**99.86%** (1,471/1,473)	**98.78%** (1,455/1,473)	**100.00% (Reference)**
1. Short Routing (293 tok, $K = 2$)	**30.01 ms** (9,763 tok/s)	32.82 ms (8,929 tok/s)	31.69 ms (9,245 tok/s)	43.23 ms (6,778 tok/s)
2. Typed Decisions (736 tok, $K = 5$)	**50.29 ms** (14,635 tok/s)	62.40 ms (11,795 tok/s)	61.80 ms (11,909 tok/s)	63.58 ms (11,576 tok/s)
3. When2Call Governance (1,068 tok, $K = 1$)	**66.06 ms** (16,167 tok/s)	87.78 ms (12,167 tok/s)	89.34 ms (11,955 tok/s)	77.77 ms (13,733 tok/s)
4. PhishNChips Security (1,303 tok, $K = 9$)	**75.86 ms** (17,176 tok/s)	108.55 ms (12,004 tok/s)	109.97 ms (11,849 tok/s)	90.32 ms (14,426 tok/s)
5. SafeAgents Judge (2,217 tok, $K = 2$)	**125.85 ms** (17,616 tok/s)	181.70 ms (12,201 tok/s)	183.50 ms (12,082 tok/s)	138.06 ms (16,058 tok/s)
6. Batch $B = 4$ Typed Decisions (2,944 tok, 20 dec)	**152.39 ms** (19,319 tok/s · 131.2 dec/s)	248.50 ms (11,847 tok/s)	245.20 ms (12,006 tok/s)	175.85 ms (16,742 tok/s · 113.7 dec/s)

First-Stage Long-Context Tool Reranking and Cascaded Reasoning Retrieval. On large tool registries (ToolRet), FastBrain-9B ranks 32 candidate tools in 672.51 ms (13.7k tok/s) at 67.26% nDCG@10. For theorem-proving and deductive retrieval workloads (BRIGHT, where 27.36B `Clef` reaches 48.03% vs. our 38.58%), FastBrain-9B serves as a fast first-stage filter on `G4 6000` that routes top-scoring candidates to a 27B+ verifier.

7 Discussion and Conclusion

Our findings show that the governance bottlenecks observed in 9B-class single-pass decision models trace to three addressable modeling and calibration factors: open-set logit inflation across high-cardinality choice spans, action-role prior bias against abstention in tool governance, and cross-task interference during uniform multi-task fine-tuning. By combining Selective Dual-Teacher KL Routing, co-adapted interpolation of all 208 backbone LoRA factor matrices and 122 `JointSchemaHead` tensors, and request-time schema-dependent post-processing, FastBrain-9B raises the unweighted ten-benchmark Decision Index mean from 74.89% to **80.55%** (+5.66 pp same-harness; +5.59 pp vs. the 74.96% published leaderboard snapshot).

Three concrete limitations define directions for future investigation. First, on reasoning-intensive document retrieval (BRIGHT; Su et al., 2025), FastBrain-9B (38.58% nDCG@10) trails the 27.36B `Clef` model (48.03%) and 27.0B `Jev` (43.37%). Because single-pass System-1 scoring executes zero intermediate chain-of-thought tokens, multi-hop deductive matching is bounded by the depth of the 32-layer prefill representation; incorporating lightweight latent recurrence loops before `JointSchemaHead` pooling is a promising extension. Second, while the single merged checkpoint with role and two-probe post-processing achieves 80.08% across the 10 benchmarks at a single-model 18.82 GB footprint, reaching the full 80.55% configuration utilizes confidence-gated specialist routing across co-trained checkpoints on API-Bank and CLINC150+OOS; distilling the routed ensemble back into a single second-generation checkpoint will unify the remaining +0.47 pp gain. Third, because the When2Call evaluation split contains zero gold `direct`-answer instances, our validation-tuned role offsets ($b_{\text{direct}} = 0.00$) are optimized for tool-governance regimes where unverified direct prose is a distractor; evaluating on mixed conversational splits containing gold `direct` answers and learning an amortizing hypernetwork that predicts role offsets directly from option-span embeddings will generalize calibration to arbitrary user-defined schemas.

In conclusion, FastBrain-9B demonstrates that a 9.41B-parameter single-pass hybrid decision model can match or surpass 27B-scale models on structured tool governance, open-set intent routing, and API selection while operating within a 24.71 GB peak GPU memory footprint at sub-100 ms amortized latency on a single NVIDIA RTX PRO 6000 Blackwell GPU.

Data and Code Availability

The complete merged `bfloat16` model weights (four shards `model-00001-of-00004.safetensors` to `00004`), the 116.1M-parameter decision head (`joint_head.safetensors`), the native `llama.cpp` GGUF quantizations (`Q4_K_M`, `Q8_0`, and `BF16`), and the inference and schema post-processing runtime (`joint_schema_model.py`) are publicly available under the Apache-2.0 License at https://huggingface.co/TypeScript/FastBrain-9B and https://github.com/uluckyXH/FastBrain. The decontaminated 108,278-instance structured decision training corpus is available at https://huggingface.co/datasets/TypeScript/FastBrain-Instruct.

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